



Data pipeline and Streamlit application for predicting employee attrition

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Problem Statement

Employee attrition is a universal problem for any business. Employees will leave a company for many reasons: better salary, mis-management, misalignment with business culture, etc. It is important for successful businesses to retain and prioritize their employees to minimize employee attrition. Companies must make sure their employees are happy with their current role, current manager, salary, etc. Losing critical talent can result in unforeseeable costs, such as: loss of productivity, employee burnout, loss of company knowledge, and money. According to SHRM, the cost of replacing an employee could be from 50% to 200% of the employee's annual salary.¹

What if we could use our Human Resource (HR) data and train a machine learning model to predict employee attrition? HR departments could use this information to focus their retention programs towards specific employees, especially those who are a great importance to the company.

Objectives

We have three objects for this project:

1. Train and compare machine learning (ML) models to determine which ML model performs the best.
2. Build a data pipeline to train and implement the best ML model.
3. Build a Streamlit application using the pre-trained ML model to predict employee attrition.

Business questions

How well can we predict employee attrition? With our HR data, we will try to build a ML model to predict employee attrition with high confidence. We will also determine what factors attribute the most to employee attrition.

¹ The myth of replaceability: Preparing for the loss of Key Employees. (n.d.).
<https://www.shrm.org/executive-network/insights/myth-replaceability-preparing-loss-key-employees>

Methodology

Data

The data is kaggle:

<https://www.kaggle.com/davidafolayan/hr-analytics?select=Employee.csv>

Two data sets:

1. Employee.csv
 - a. 23 columns and 1470 rows

i. EmployeeID	xii. EducationField
ii. FirstName	xiii. JobRole
iii. LastName	xiv. MaritalStatus
iv. Gender	xv. Salary
v. Age	xvi. StockOptionLevel
vi. BusinessTravel	xvii. OverTime
vii. Department	xviii. HireDate
viii. DistanceFromHome (KM)	xix. Attrition
ix. State	xx. YearsAtCompany
x. Ethnicity	xxi. YearsInMostRecentRole
xi. Education	xxii. YearsSinceLastPromotion
	xxiii. YearsWithCurrManager
2. PerformanceRating.csv
 - a. 11 columns and 6708 rows

i. PerformanceID
ii. EmployeeID
iii. ReviewDate
iv. EnvironmentSatisfaction
v. JobSatisfaction
vi. RelationshipSatisfaction
vii. TrainingOpportunitiesWithinYear
viii. TrainingOpportunitiesTaken
ix. WorkLifeBalance
x. SelfRating
xi. ManagerRating

Data Preparation

We cleaned and prepared the data before training the ML models. We conducted the following cleaning and preparation steps:

1. For the performance rating data, we only kept the latest performance ratings for each employee.
2. Merged the two datasets together using EmployeeID and only keeping the rows where EmployeeID was present in both datasets (i.e., inner join).
3. Removed unnecessary columns.
 - a. EmployeeID
 - b. FirstName
 - c. LastName
 - d. Index
 - e. PerformanceID
4. After merging and cleaning, the dataset had 1280 rows and 30 columns.

Exploratory Data Analysis (EDA)

Here are the important data trends we found after EDA.

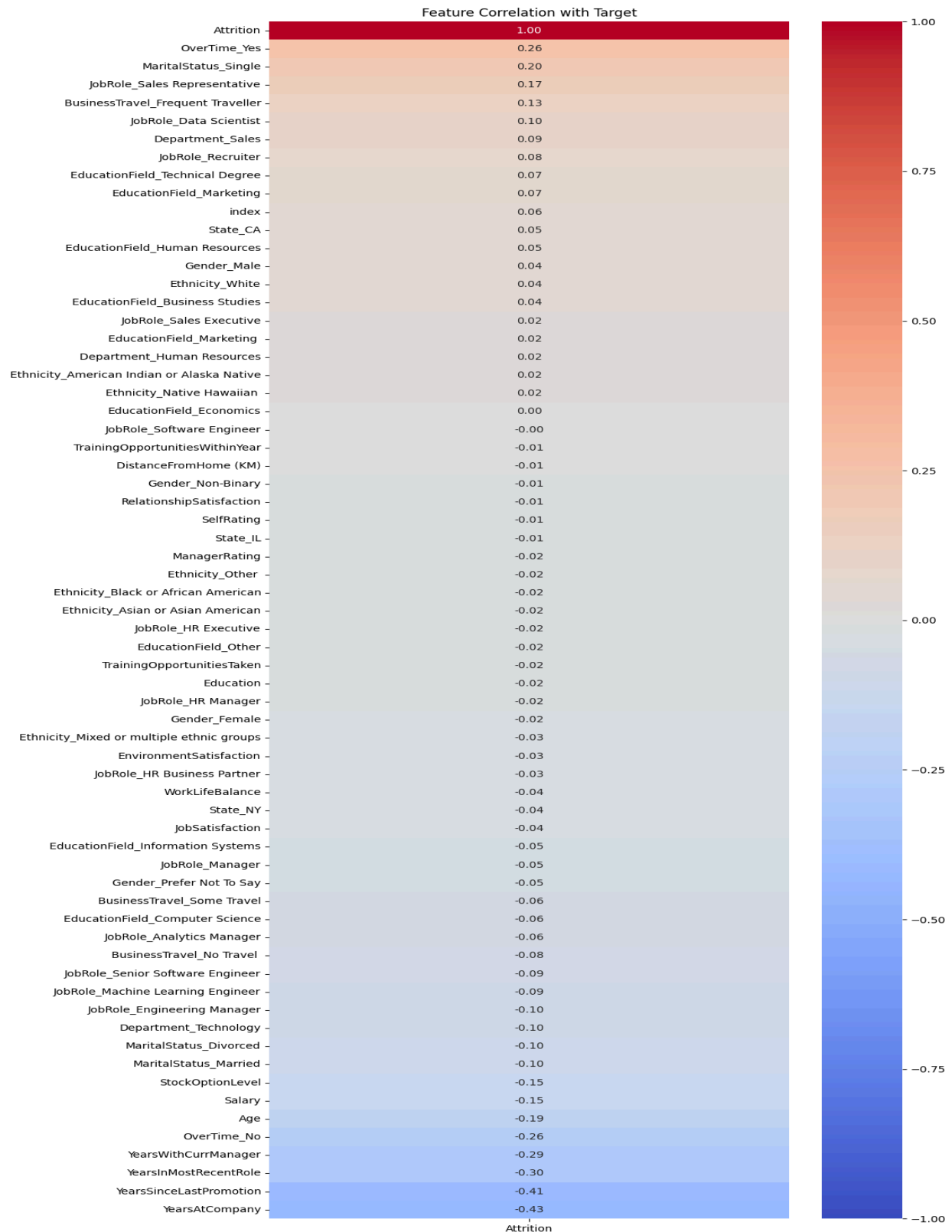
Total Attrition

Attrition status	Number of Employees	Attrition Rate
Stays	1043	81.5%
Leaves	237	18.5%

Attrition by Department

Department	Total_Employees	Number of Employees who left	Attrition_Rate
Human Resources	53	12	23%
Sales	384	92	24%
Technology	843	133	16%

Feature Correlation with Target



Top 3 positive correlations with attrition:

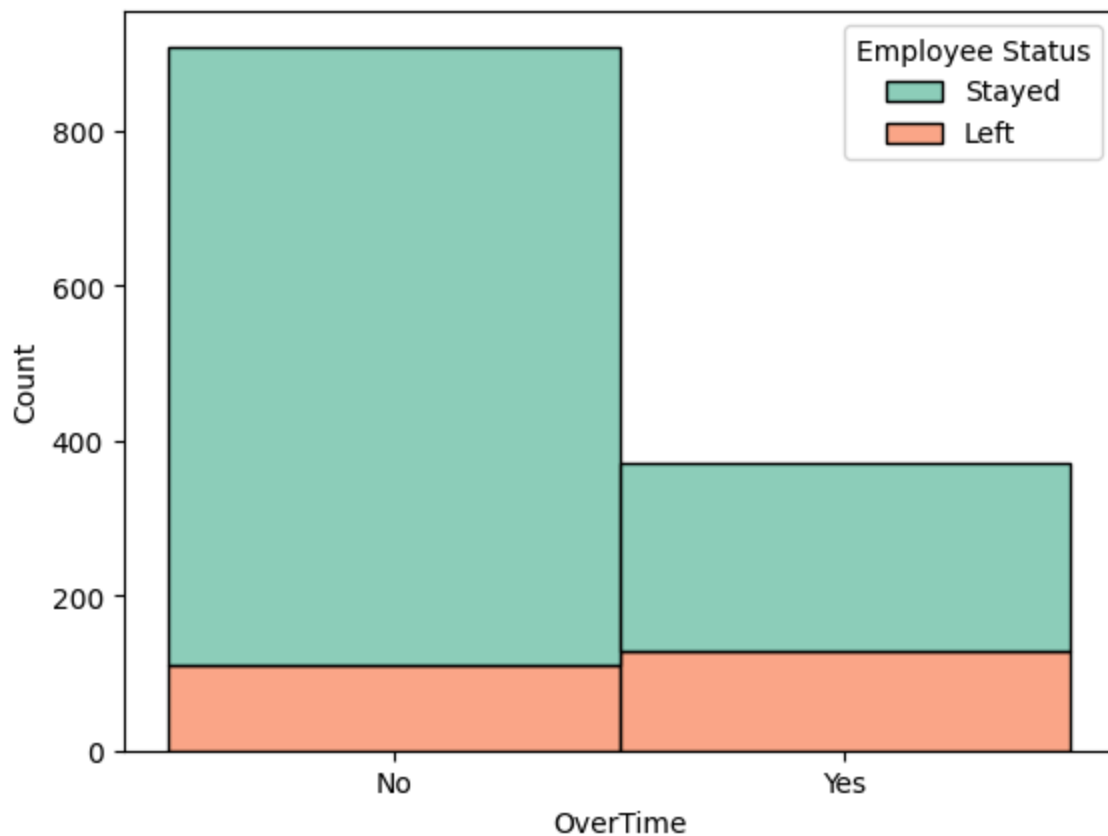
1. Overtime_Yes: 0.26
2. MaritalStatus_Single: 0.20
3. JobRole_Sales Representative: 0.17

Top 3 negative correlations with attrition:

1. YearsAtCompany: -0.43
2. YearsSinceLastPromotion: -0.41
3. YearsinMostRecentRole: -0.30

Histograms & Violin Plots of Attrition by Highly Correlated Features

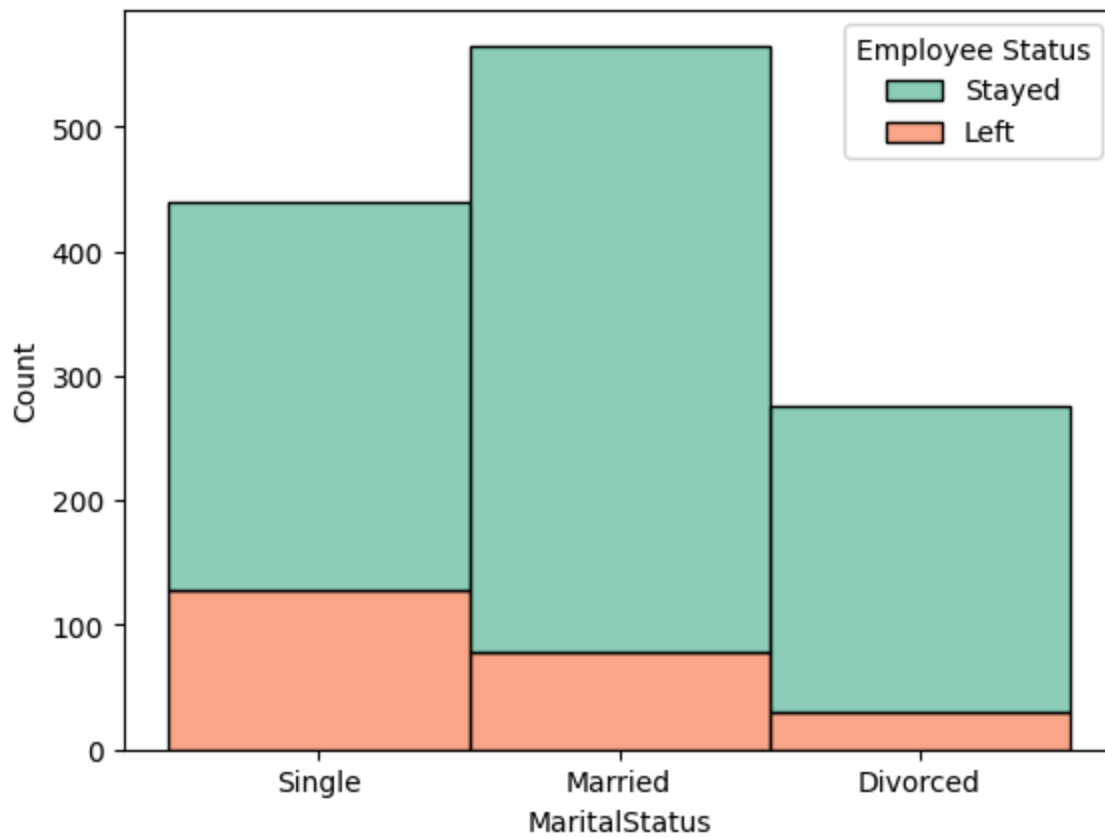
Histograms of Attrition by Overtime



The attrition rate for employees who do not work over time is 12%.

The attrition rate for employees who work over time is 34%.

Histograms of Attrition by MaritalStatus

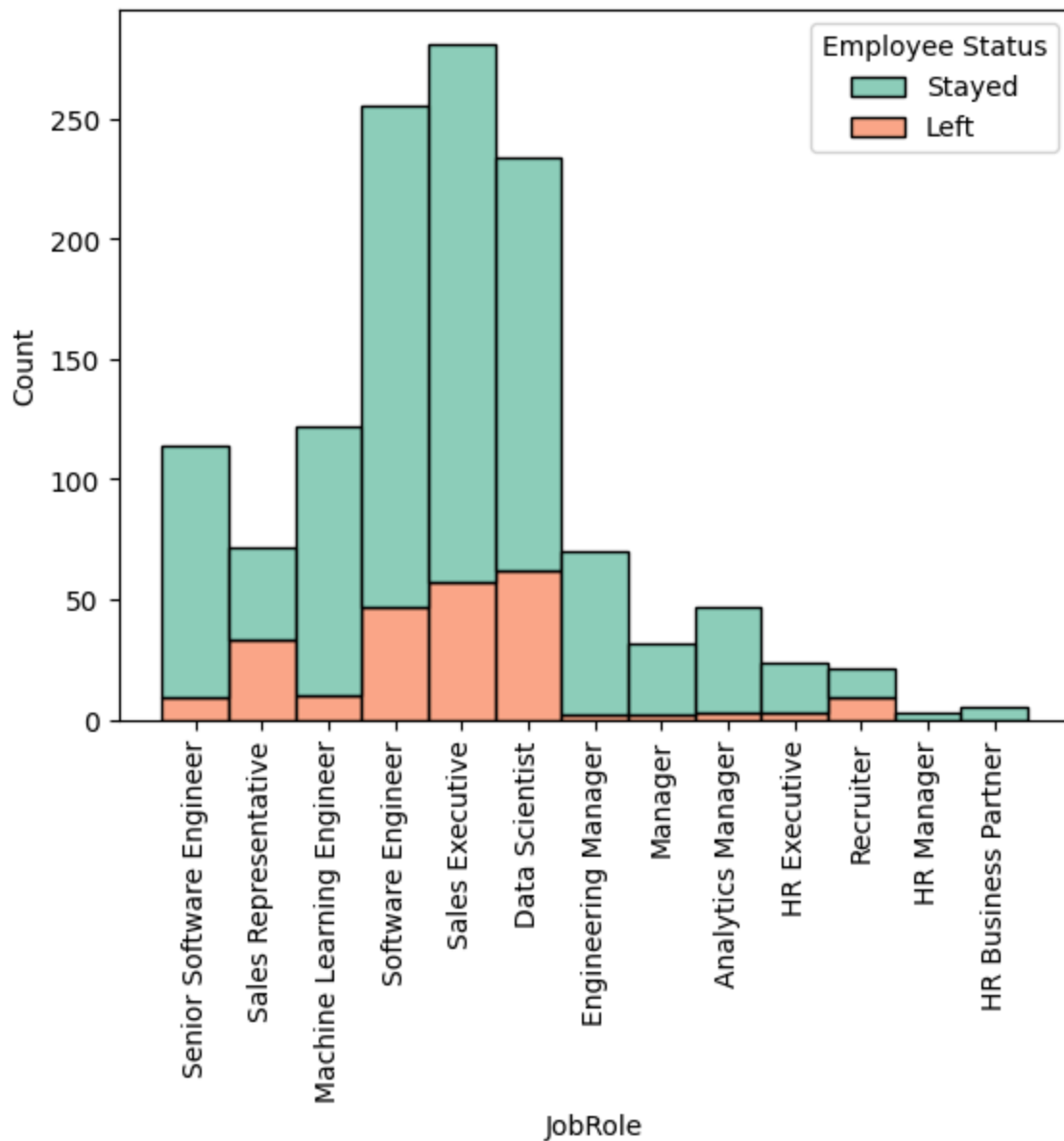


The attrition rate for employees who are single is 29%.

The attrition rate for employees who are married is 14%.

The attrition rate for employees who are divorced is 11%.

Histograms of Attrition by JobRole

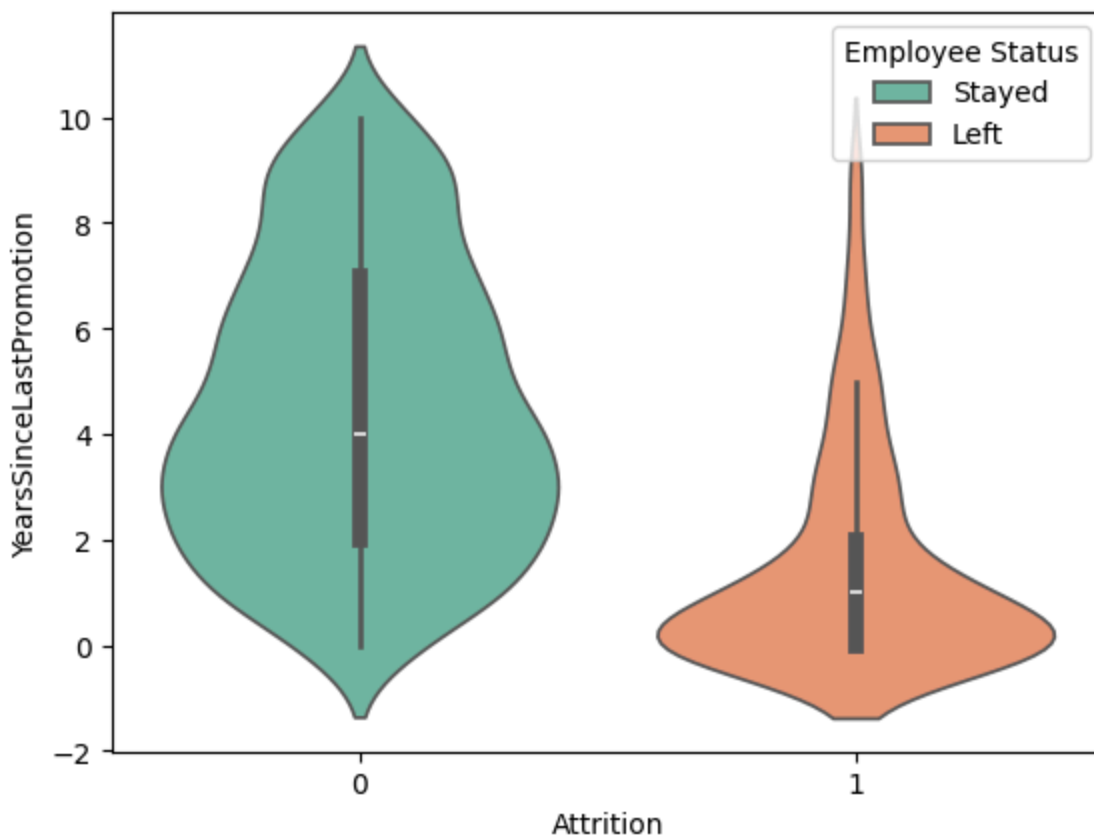


Sales Representatives have the highest attrition rate followed by Recruiter and Data Scientist. Below is a table showing the attrition rate by job role.

Table of attrition rate by job role:

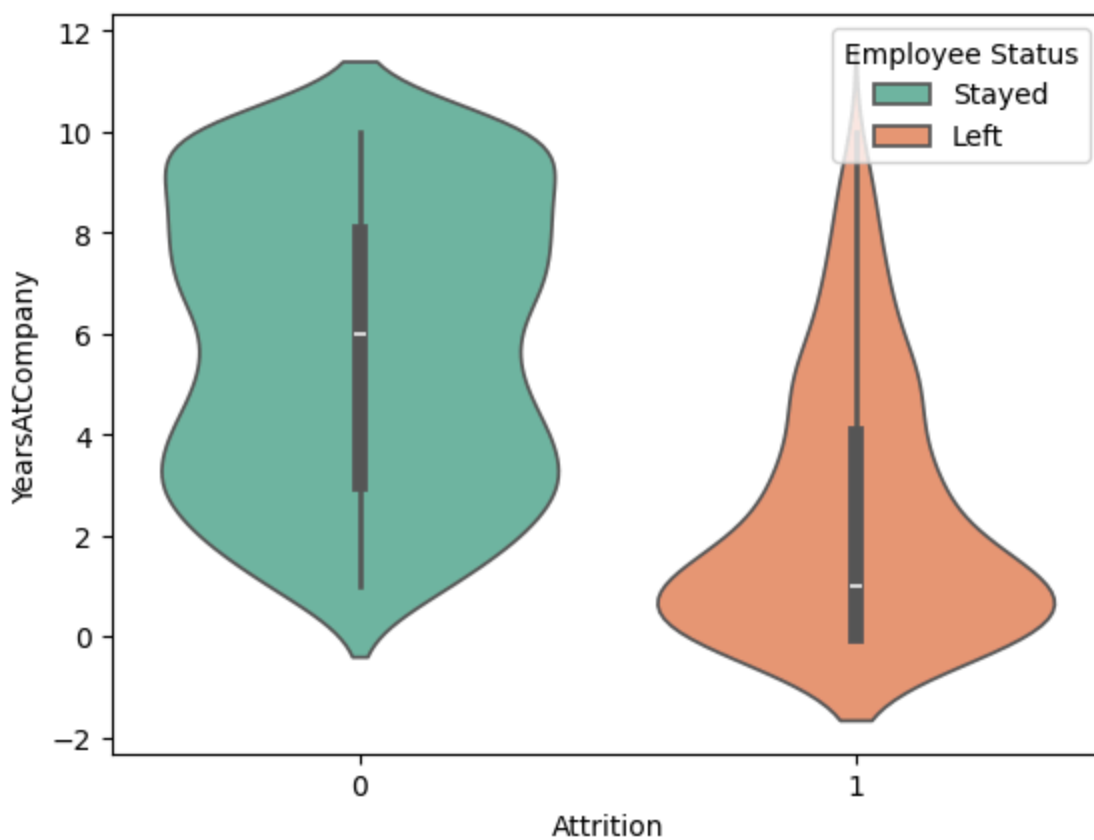
JobRole	Attrition Rate
Sales Representative	46%
Recruiter	43%
Data Scientist	26%
Sales Executive	20%
Software Engineer	18%
HR Executive	13%
Machine Learning Engineer	8%
Senior Software Engineer	8%
Analytics Manager	6%
Manager	6%
Engineering Manager	3%
HR Business Partner	0%
HR Manager	0%

Violin Plots of Attrition by YearsSinceLastPromotion



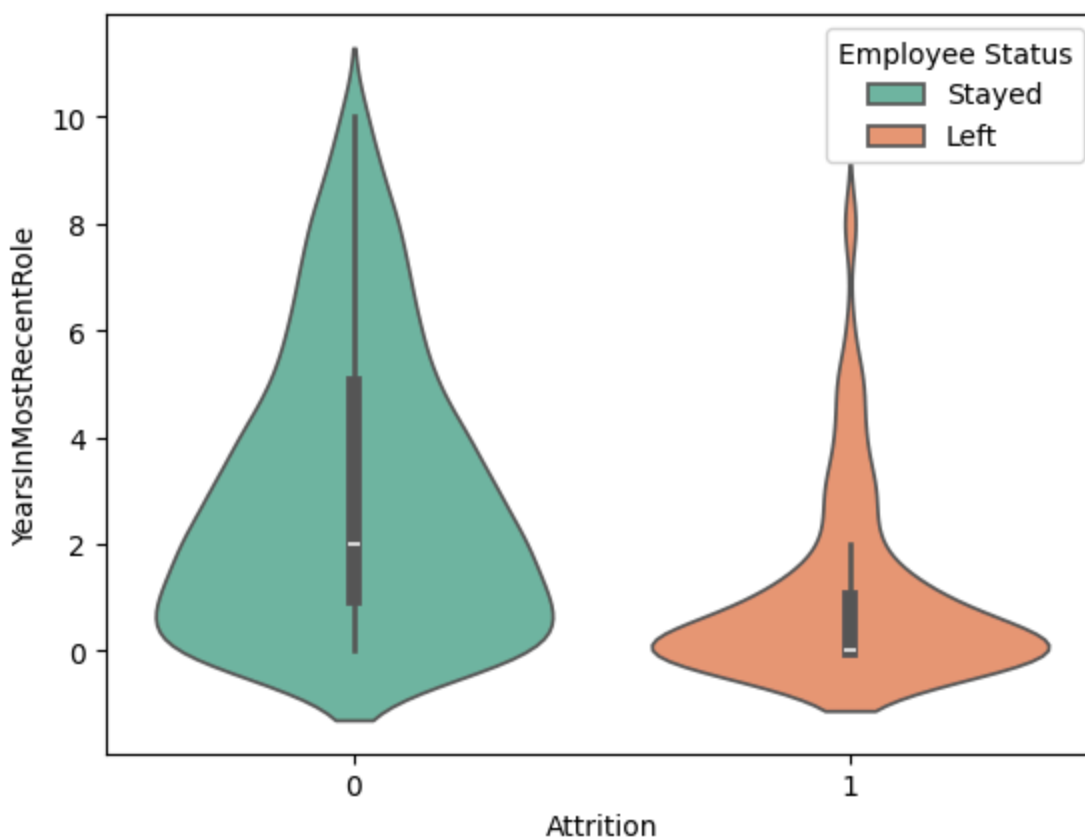
The majority of the employees leave the company if they do not receive a promotion within the first two years.

Violin Plots of Attrition by YearsAtCompany



Most employees leave the company within the first three years.

Violin Plots of Attrition by YearsInMostRecentRole



Most employees will leave the company within the first two years of being in the same role.

Modeling

We used the cleaned HR data to train Machine Learning models to predict employee attrition. We compared several different models to see which model performs the best. Here are the ML models we trained and compared:

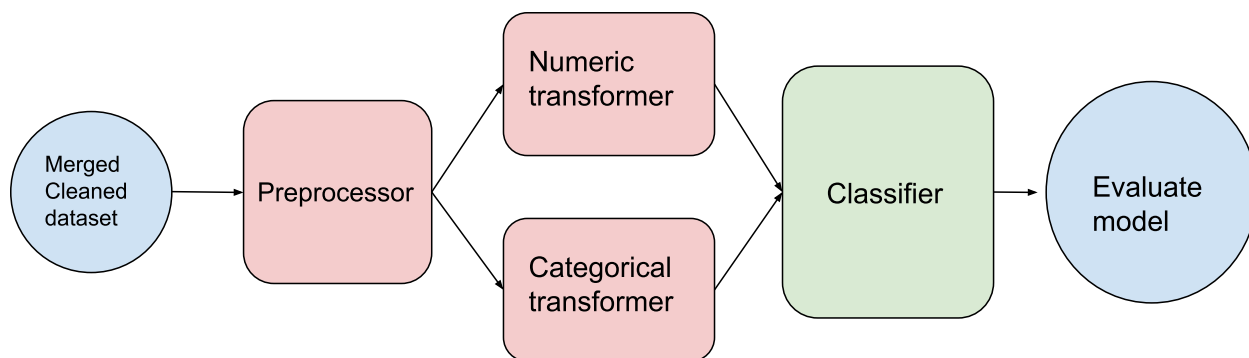
- Logistic Regression
- Random Forest
- Support Vector Machine (SVC)
- Adaptive Boosting
- K-Nearest Neighbors (KNN)
- Gaussian Naive Bayes
- Gradient Boosting Classifier
- Bootstrap Aggregating (Bagging)

A data pipeline was used to streamline the data wrangling to be able to train, fit, and compare the ML models.

The data pipeline process:

1. Input merged cleaned data
2. Preprocessor
 - a. Numeric columns go through a numeric transformer
 - i. Imputes missing values with median
 - ii. Scales values: creates z-scores where mean = 0 and standard deviation = 1
 - b. Categorical columns go through a categorical transformer
 - i. Imputes missing values with mode
 - ii. Encodes categories to dummy variables
3. Classifier
 - a. ML models
4. Evaluate ML models

Data pipeline diagram:



Outcomes

Accuracy, Precision, Recall, and F1-Score Table

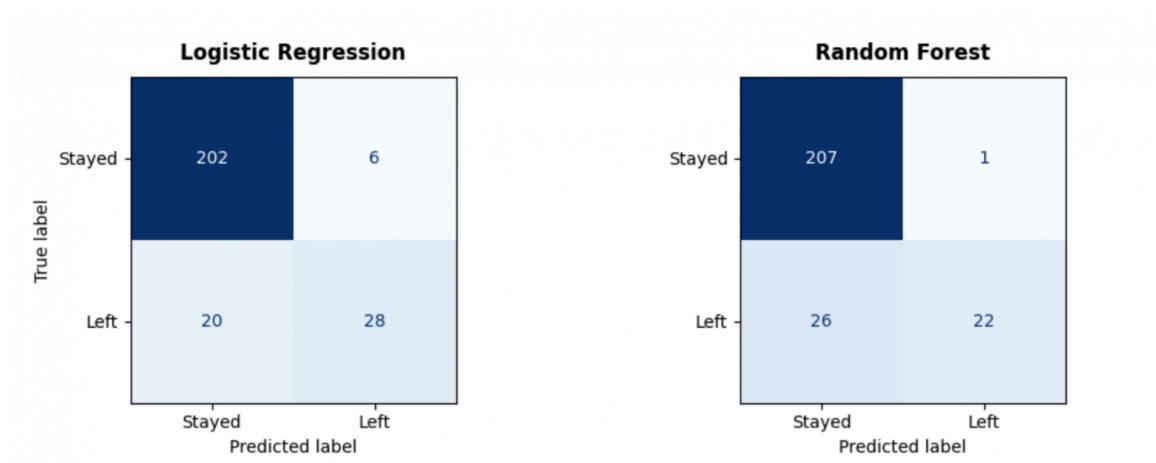
ML Algorithms ▾	#	Accuracy ▾	#	Precision ▾	#	Recall ▾	#	F1-Score ▾
Logistic Regression		0.898		0.824		0.583		0.683
Support Vector Machine		0.898		0.867		0.542		0.667
Bagging		0.898		0.923		0.500		0.649
AdaBoost		0.891		0.833		0.521		0.641
GBC		0.891		0.833		0.521		0.641
Random Forest		0.895		0.957		0.458		0.620
KNN		0.840		0.578		0.542		0.559
GNB		0.336		0.220		1.000		0.361

Logistic Regression, Support Vector, and Adaptive Bagging all had an accuracy score of 89.8%. However, with our imbalanced dataset with only 18.5% of the sample that leave the company, accuracy may not be the best metric to evaluate our models. Given, any model that always predicts an employee will stay with the company will be correct about 82% of the time.

Precision may be a better metric to evaluate our ML models, as it measures the accuracy of positive predictions. In other words, the accuracy of predicting employees actually leaving the company.

If the business's focus is on correctly predicting who will leave, then the Random Forest model would be the best model to make predictions. If the business wanted to lower their false negatives or incorrectly predict an employee will stay but the employee actually leaves, then Logistic Regression would be more beneficial without losing much precision.

Confusion Matrices for Logistic Regression and Random Forest:



Implementation

We opted to use Random Forest as our ML model. We valued having high precision to have high confidence for predicting employee attrition. Random forest has a low recall score which means it has a higher tendency to predict employees will stay but in reality employees will leave.

We built another data pipeline that only used Random Forest as the sole classifier. The data pipeline was used to build a Streamlit application for the company. The company will be able to enter the HR data of the employee to predict employee attrition. The application will also provide a confidence level of each prediction.

Screenshots of Streamlit application:

Deploy

Attrition Prediction App

Predict Employee Attrition

Predict

Will employee leave?

Prediction: Yes

Confidence: 57.0%

HR Data Input Features

EnvironmentSatisfaction

5

JobSatisfaction

3

RelationshipSatisfaction

3

TrainingOpportunitiesWithinYear

3

TrainingOpportunitiesTaken

2

WorkLifeBalance

3

SelfRating

2

ManagerRating

2

Gender

Male

Age

Deploy

Attrition Prediction App

Predict Employee Attrition

Predict

Will employee leave?

Prediction: Yes

Confidence: 57.0%

HR Data Input Features

EnvironmentSatisfaction

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JobSatisfaction

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TrainingOpportunitiesTaken

2

WorkLifeBalance

3

SelfRating

2

ManagerRating

2

Gender

Male

Age

Business Answer

Our ML model, Random Forest, can predict employee attrition with 96% precision. However, the recall is low, 46%, meaning the model will predict employees will stay but will actually leave at a higher rate.

The company can invest in retention programs that focuses on the following:

- Career development planning
- Mentoring
- Upskill training

The retention program will help keep employees satisfied. It is important for the company to focus the program on new employees (0-2 years). Employees are less likely to leave the company after being with the company for 2 or more years.

The company can utilize the Streamlit application to predict employee attrition. It will be most beneficial for making predictions for critical employees, especially C-suite employees. It is important to know that valuable employees will stay with the company and prevent the cost of replacing valuable employees.

References

Python Libraries:

Python Libraries
import pandas as pd
import numpy as np
%matplotlib inline
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import AdaBoostClassifier
from sklearn.ensemble import GradientBoostingClassifier
from sklearn.svm import SVC
from sklearn.naive_bayes import GaussianNB
from sklearn.neighbors import KNeighborsClassifier
from sklearn.ensemble import BaggingClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
import joblib
import streamlit as st
import pickle

Codebooks:

EDA: Capstone_Ae_Sengmavong.ipynb

Data pipeline: HR_Data_Pipeline.ipynb

Trained Random Forest model: Attrition_model.pkl

Streamlit application: HR_Attrition_predict.py